Data Science and Artificial Intelligence

Machine Learning

Linear Discriminant Analysis

Lecture No. 4

By- SIDDHARTH SABHARWAL SIR

Recap of Previous Lecture

- Topic LDA
- Topic PCA
- Topic
- Topic
- Topic

Topics to be Covered

- Topic PCA
- Topic Numerical.
- Topic
- Topic
- Topic

LIFE IS LIKE
a camera
JUST FOCUS
ON WHAT IS IMPORTANT
CAPTURE
THE GOOD TIMES
DEVELOP
FROM THE NEGATIVES
IF THINGS DON'T WORK OUT
take another shot!

REVISION

LDA

PCA → Unsupervised learning

→ new principal components

→ such that they capture the max variance of data

→ new PC are linear combo of original features

→ all PC are orthogonal from eachother

What is PCA

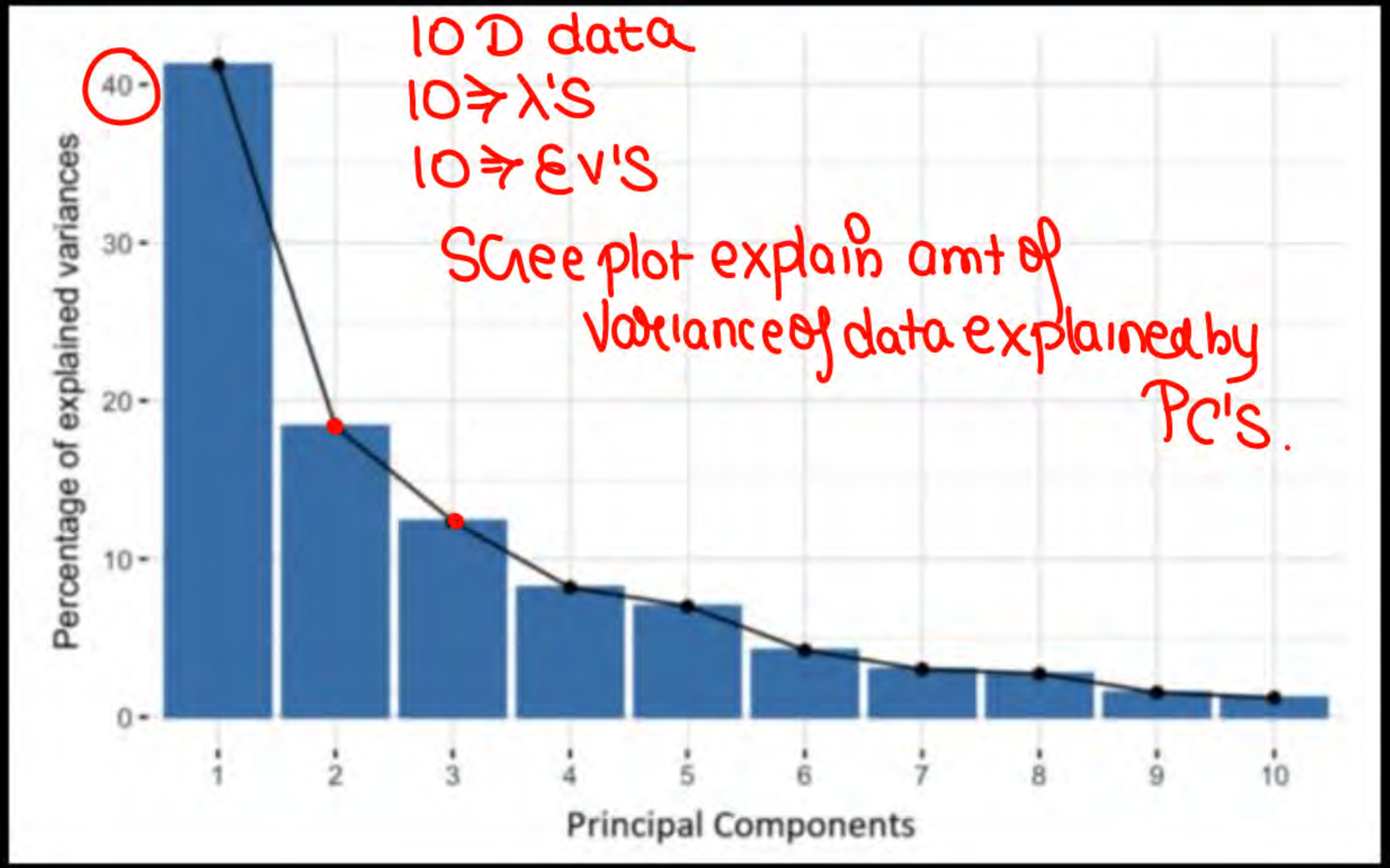
- Principal Component Analysis (PCA) is a technique for dimensionality reduction that identifies a set of orthogonal axes, called principal components, that capture the maximum variance in the data. The principal components are linear combinations of the original variables in the dataset and are ordered in decreasing order of importance. The total variance captured by all the principal components is equal to the total variance in the original dataset.
- The first principal component captures the most variation in the data, but the second principal component captures the maximum variance that is orthogonal to the first principal component, and so on.

What is PCA

- Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables.
- These combinations are done in such a way that the new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is squeezed or compressed into the first components.
- So, the idea is 10-dimensional data gives you 10 principal components, but PCA tries to put maximum possible information in the first component, then maximum remaining information in the second and so on, until having something like shown in the scree plot below.

What is PCA

Scree Plot



10D data
10 ⇒ λ's
10 ⇒ EV's

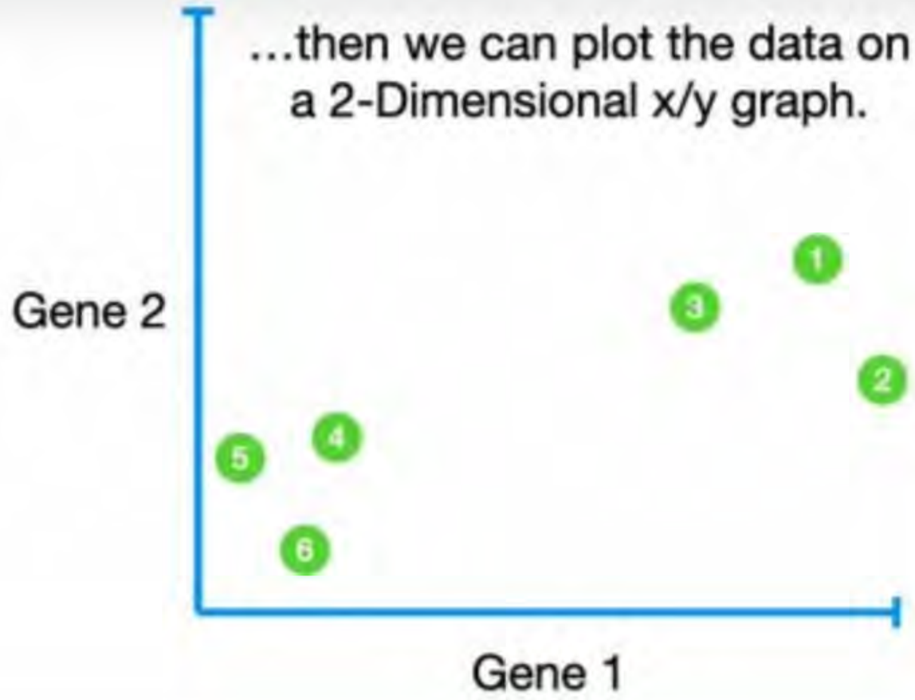
Scree plot explain amt of variance of data explained by PC's.

What is PCA

We can see that points 1,2,3 and 4,5,6, make clusters

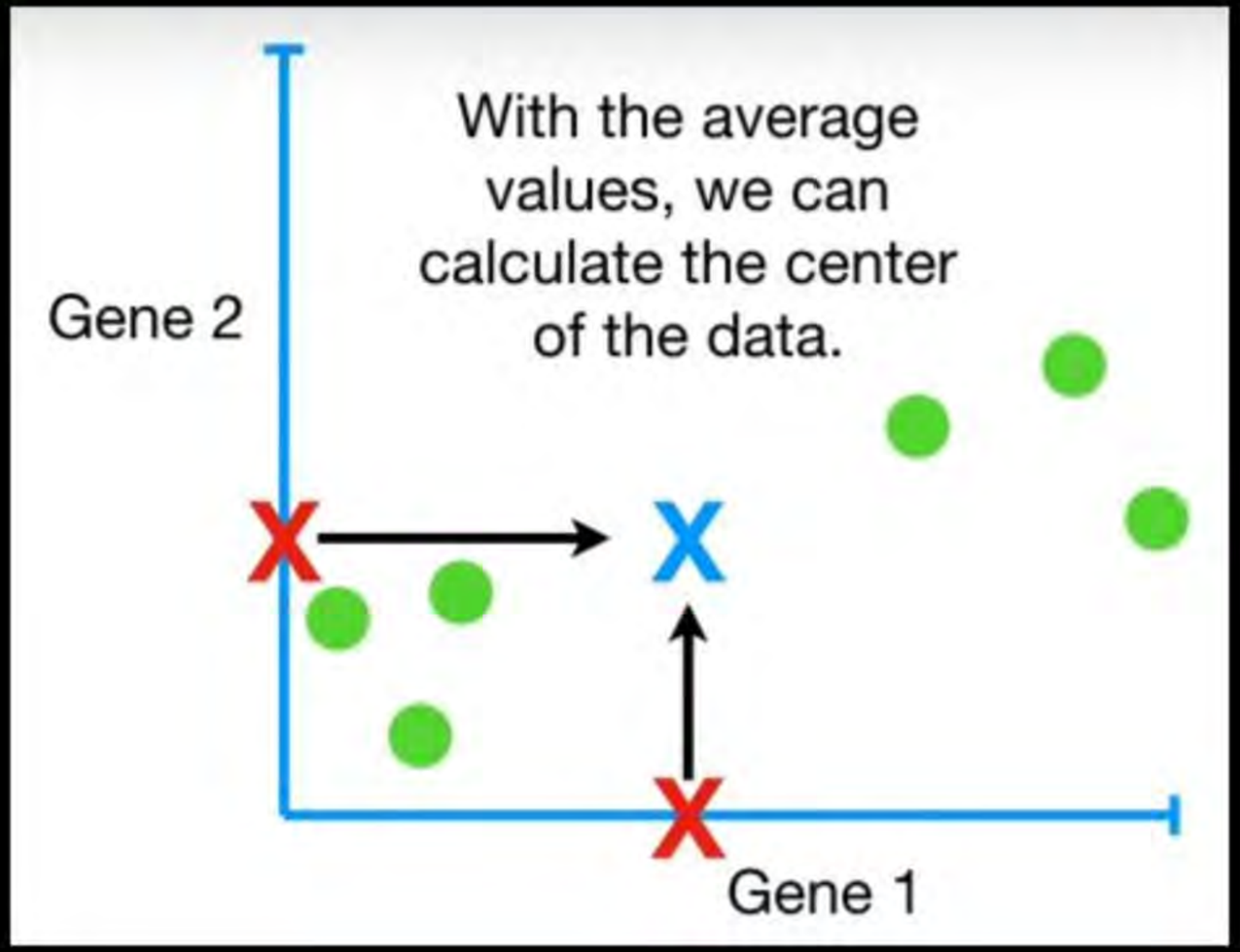
	Mouse 1	Mouse 2	Mouse 3	Mouse 4	Mouse 5	Mouse 6
Gene 1	10	11	8	3	1	2
Gene 2	6	4	5	3	2.8	1

...then we can plot the data on a 2-Dimensional x/y graph.



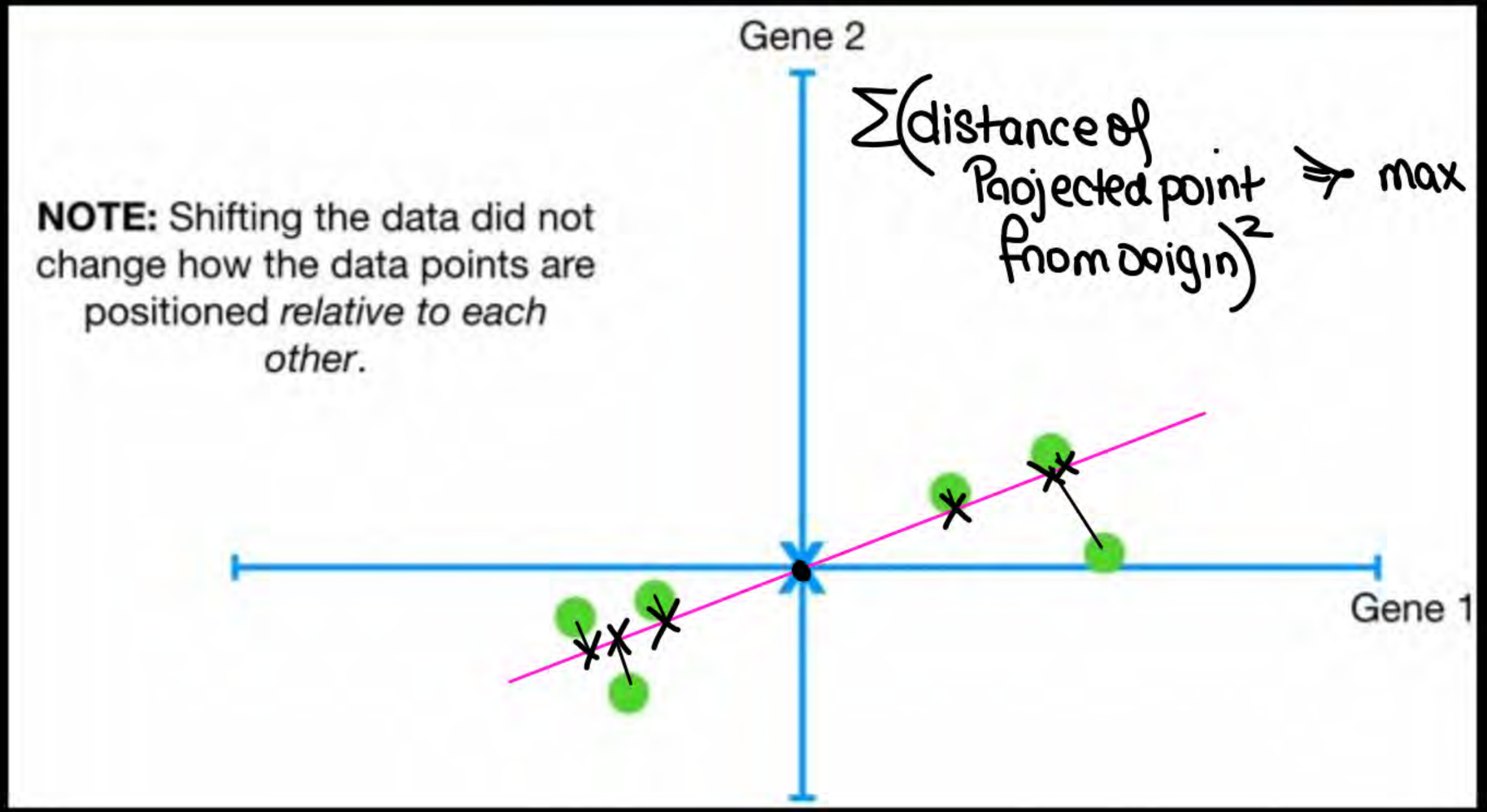
What is PCA

First of all we calculate the centre of the data



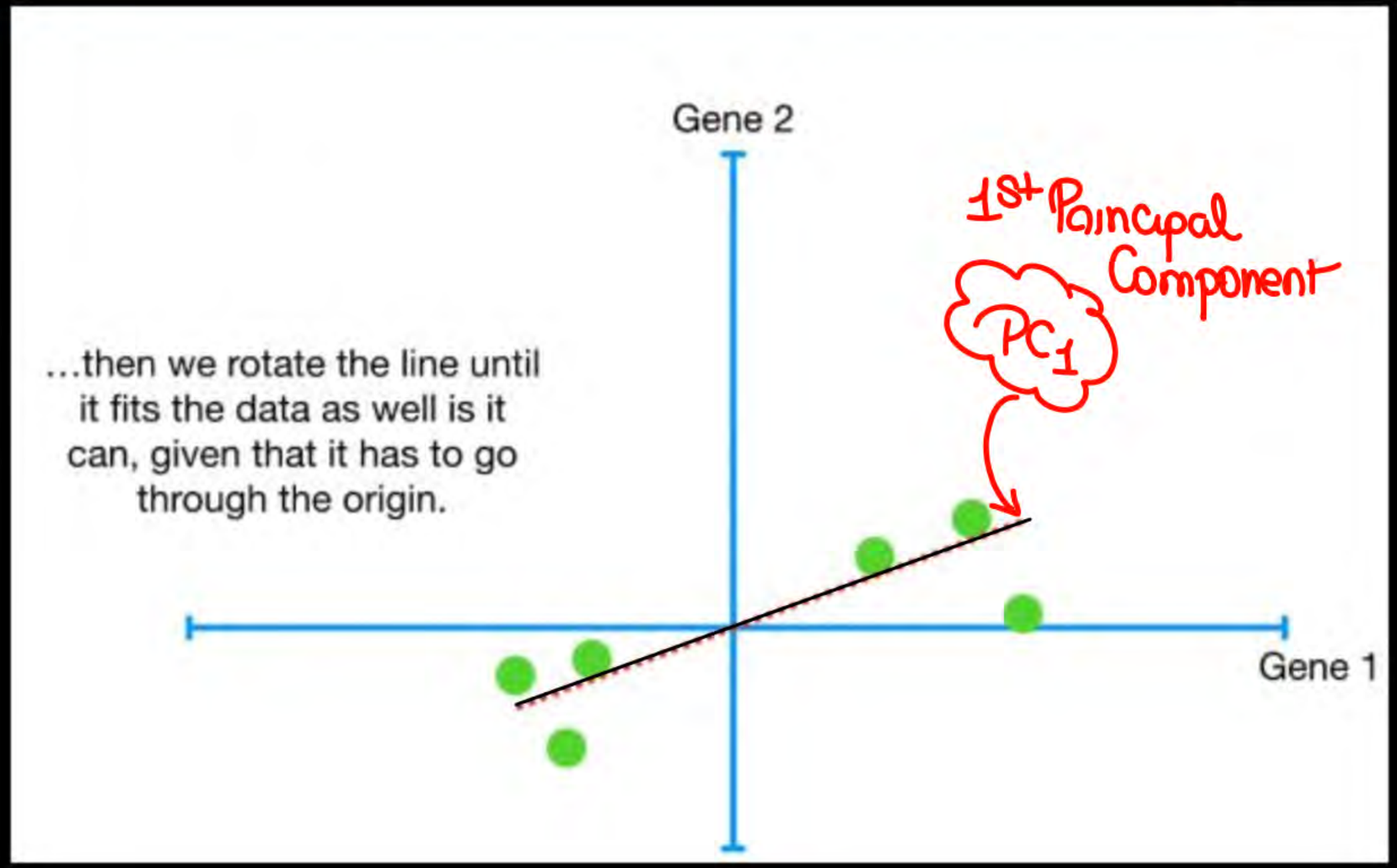
What is PCA

We shift the date to make its centre at the origin….



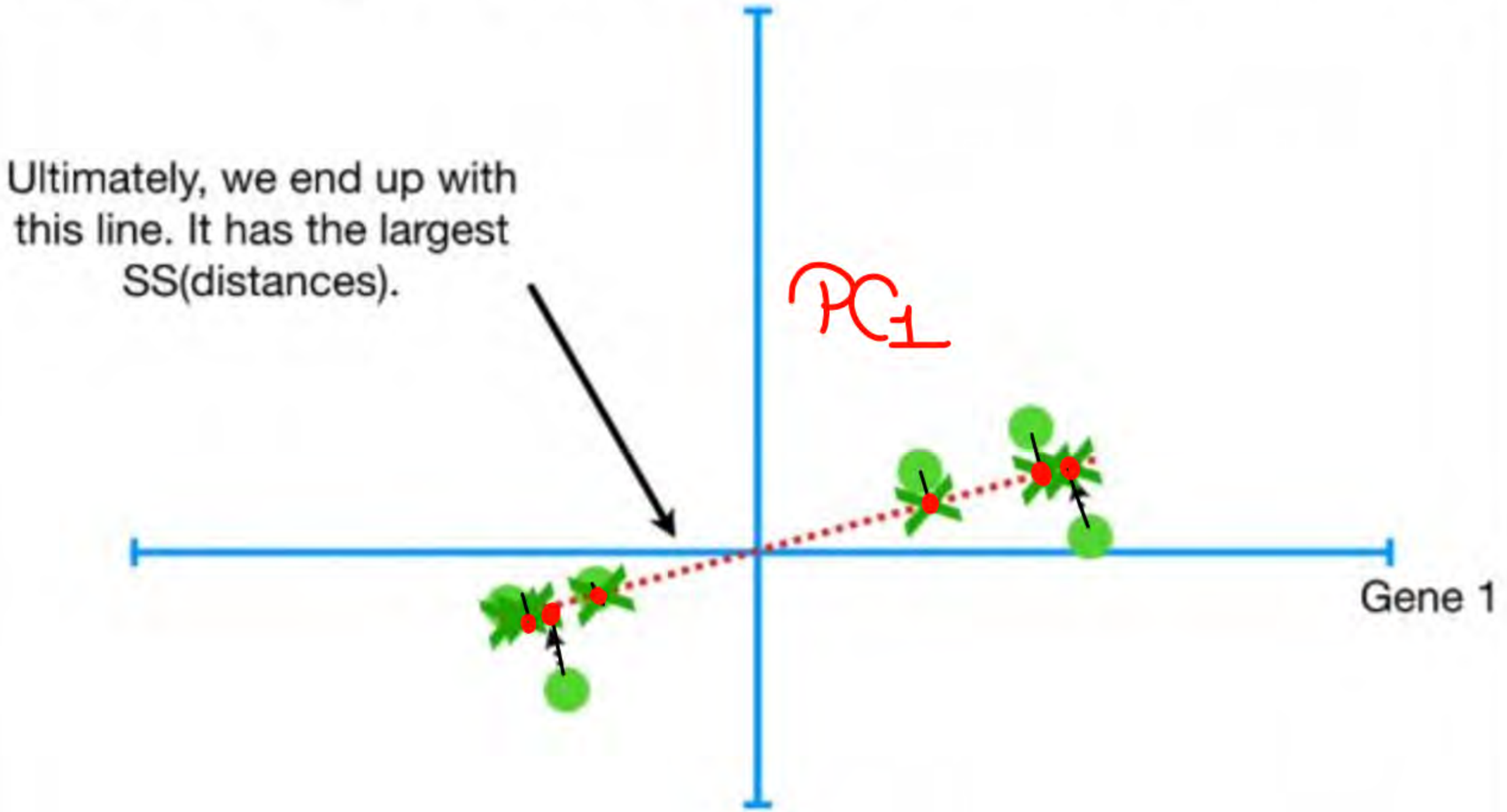
What is PCA

Now the PCA try to fit a line to the data…
Thus we need a line which has the minimum distance of point from the line or….



What is PCA

$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

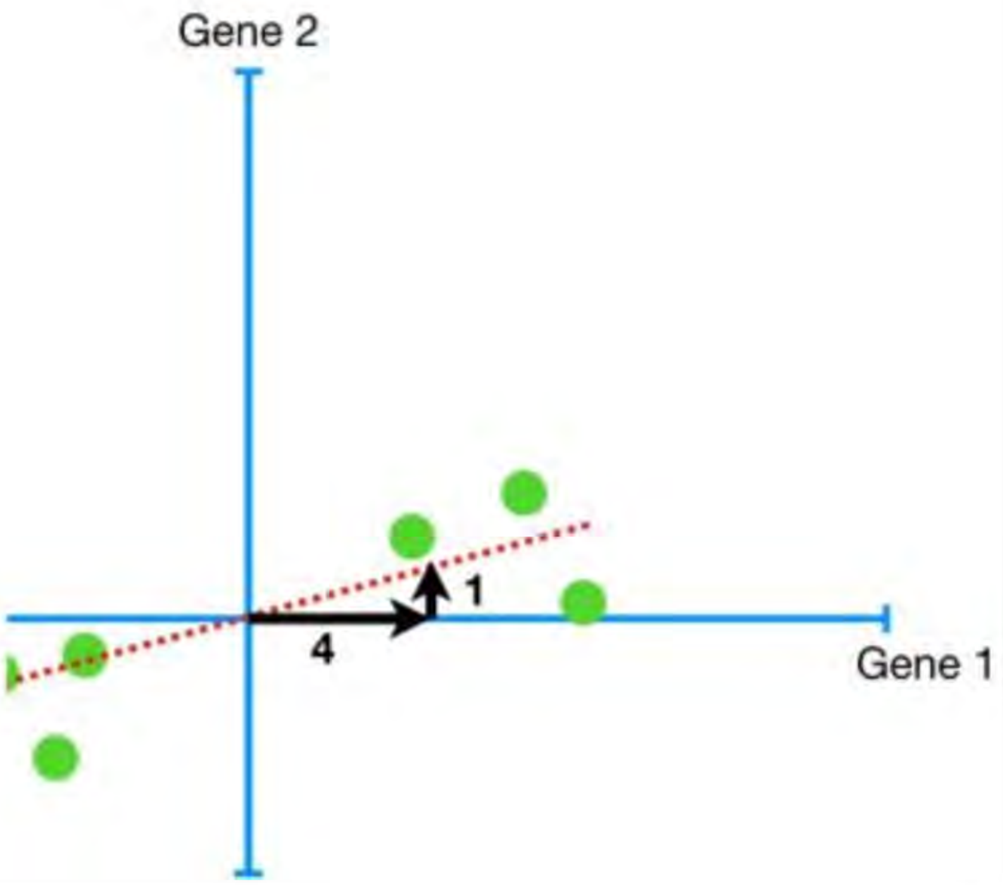


What is PCA

Skip

To make PC1
Mix **4** parts Gene 1
with **1** part Gene 2

Terminology Alert!!!!
Mathematicians call this cocktail recipe a "*linear combination*" of Genes 1 and 2.



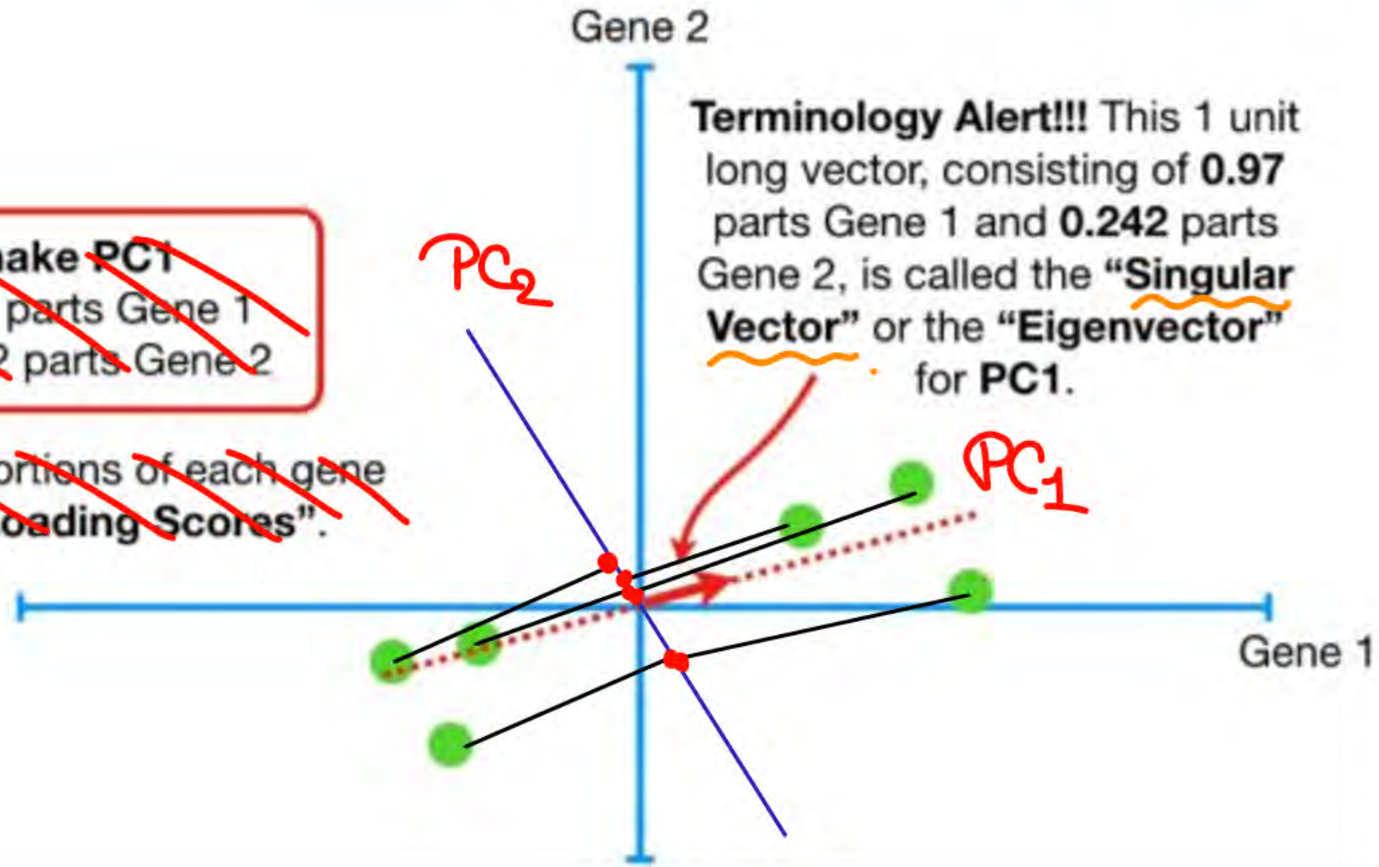
What is PCA

Let $PC_1 \Rightarrow 3x^1 + 2x^2 = 0$

$$\frac{3x^1 + 2x^2 = 0}{\sqrt{9+4}}$$

~~To make PC1~~
~~Mix 0.97 parts Gene 1~~
~~with 0.242 parts Gene 2~~

~~...and the proportions of each gene~~
~~are called "Loading Scores".~~

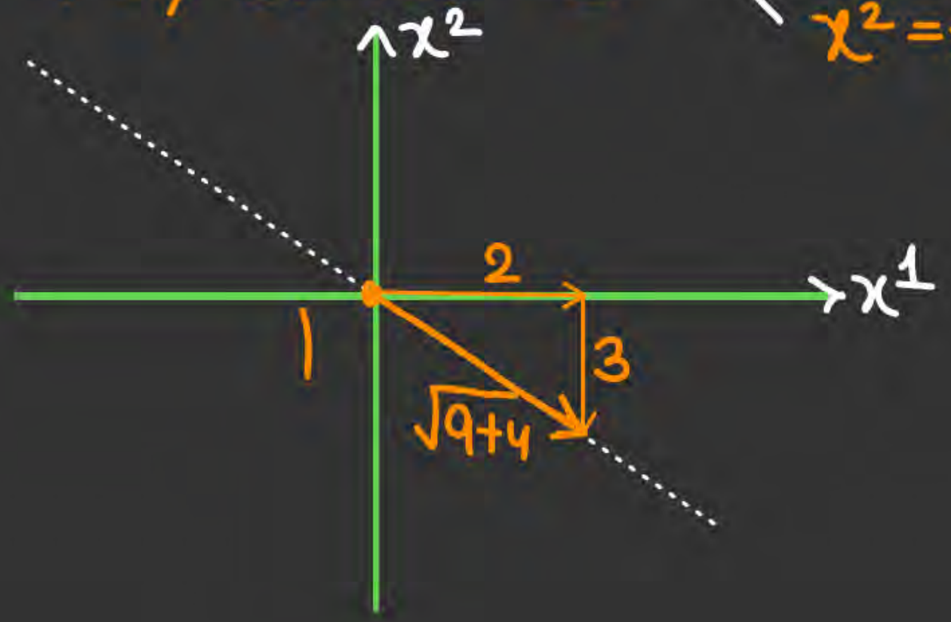


Terminology Alert!!! This 1 unit long vector, consisting of **0.97** parts Gene 1 and **0.242** parts Gene 2, is called the **"Singular Vector"** or the **"Eigenvector"** for **PC1**.

$PC_1 \Rightarrow 3x^1 + 2x^2 = 0$

$$\left(2x^2 = -3x^1\right)$$

$$x^2 = -\frac{3}{2}x_1$$



What is PCA

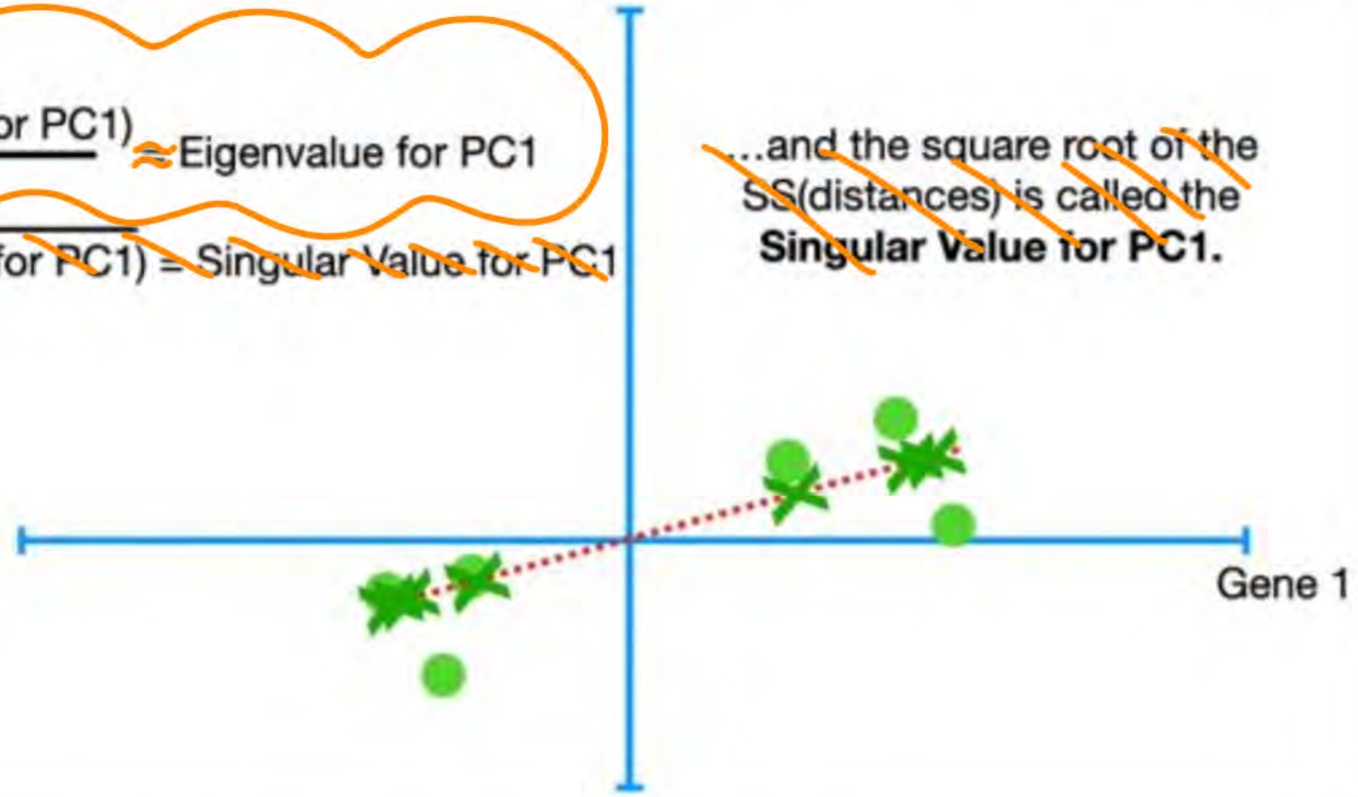
What this eigen value shows …

$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 + d_6^2$ = sum of squared distances = SS(distances)

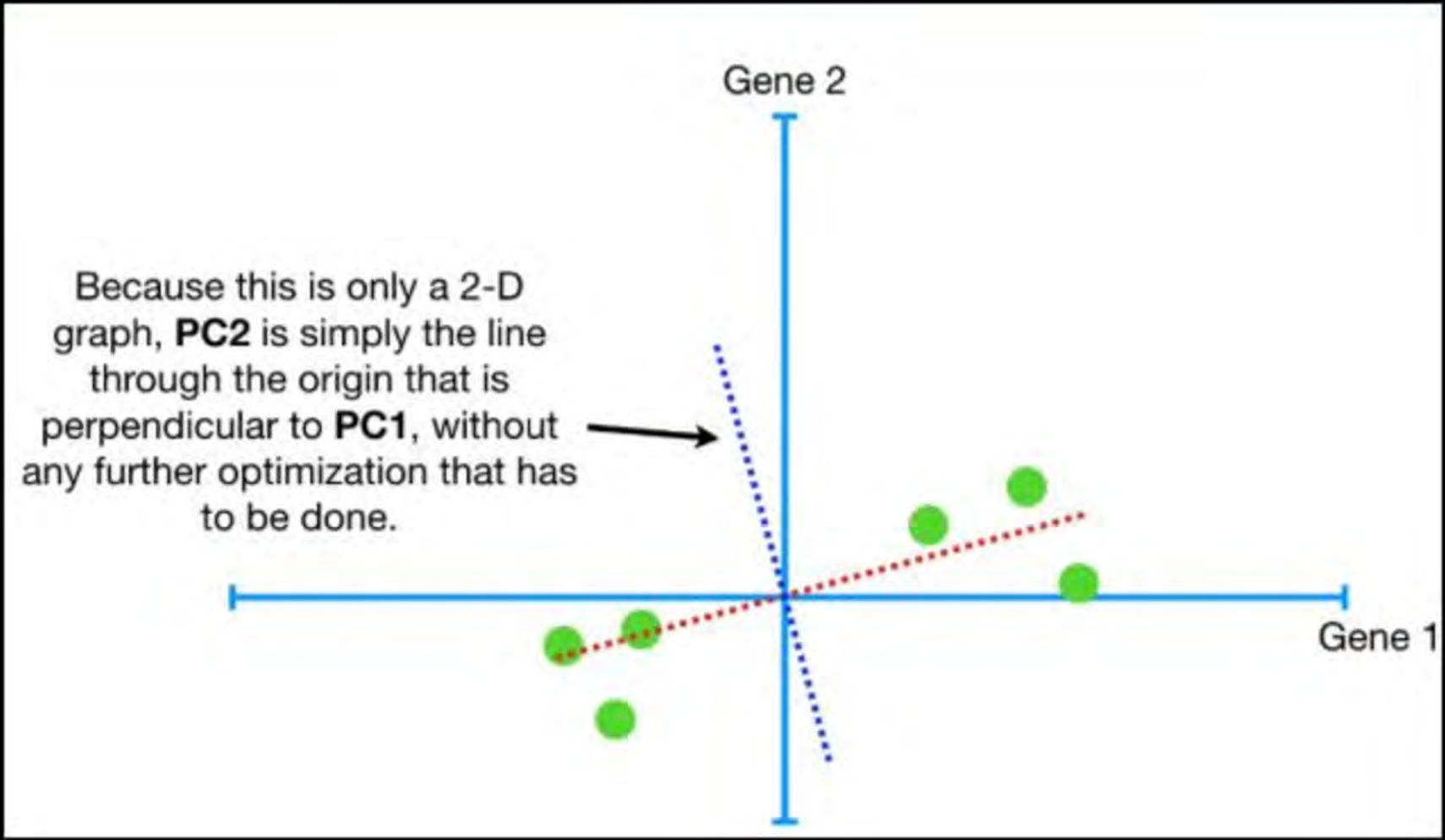
$$\frac{\text{SS(distances for PC1)}}{n-1} \approx \text{Eigenvalue for PC1}$$

$$\sqrt{\text{SS(distances for PC1)}} = \text{Singular Value for PC1}$$

…and the square root of the SS(distances) is called the **Singular Value for PC1.**



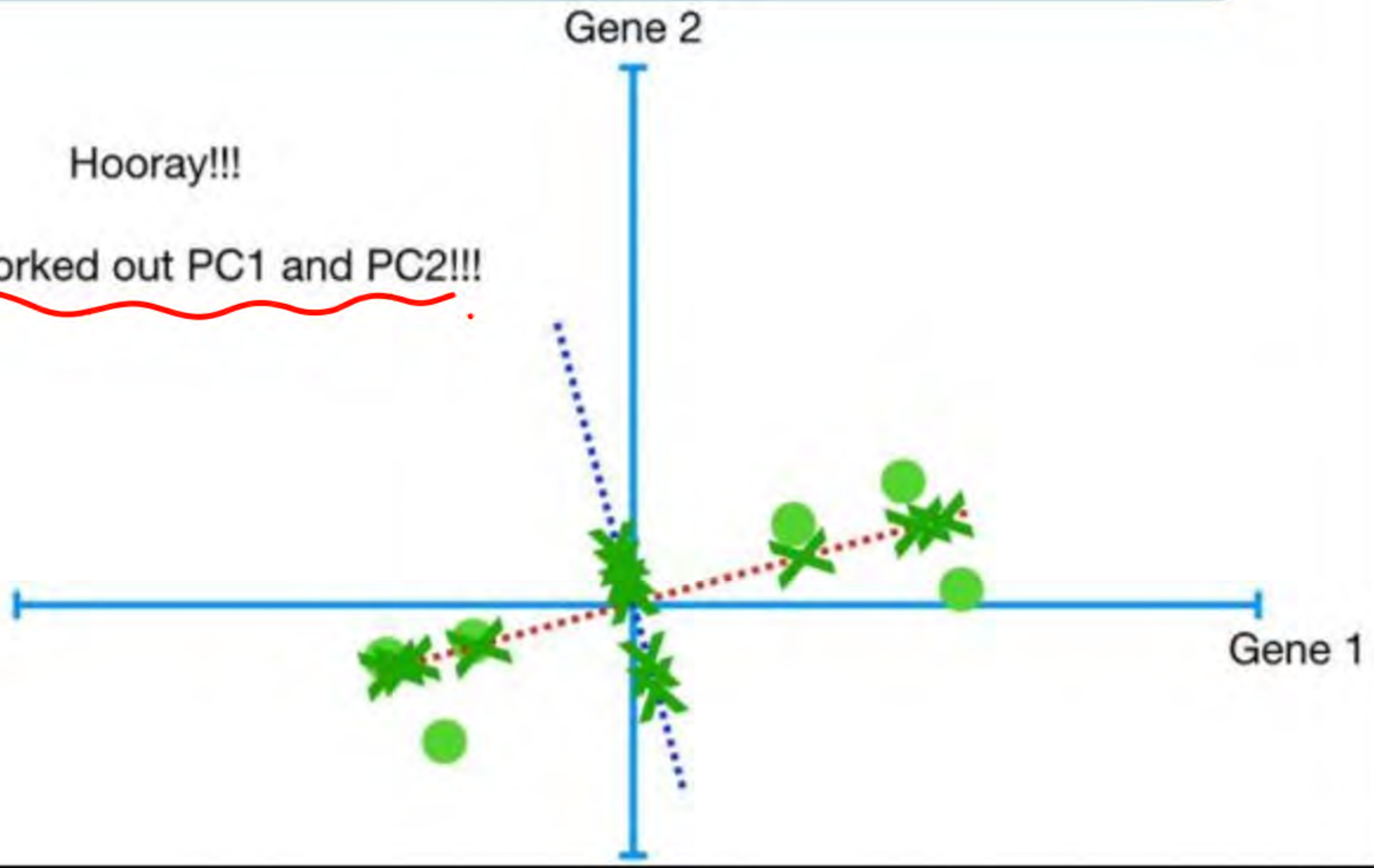
What is PCA


Gene 2
Because this is only a 2-D graph, **PC2** is simply the line through the origin that is perpendicular to **PC1**, without any further optimization that has to be done.
Gene 1

What is PCA

Hooray!!!

We've worked out PC1 and PC2!!!



So steps to create PCA ⇒

Step 1 ⇒ Mean of data = 0

Variance = 1

$$u_1 = \frac{\sum x_i^1}{N}$$

σ_2

$$\begin{bmatrix} x_1^1 & x_2^1 & | & x_N^1 \\ x_1^2 & x_2^2 & | & x_N^2 \\ \vdots & \vdots & | & \vdots \\ x_1^D & x_2^D & | & x_N^D \end{bmatrix}$$

u_2

$$\Rightarrow \sigma_1^2 = \frac{\sum (x_i^1 - u_1)^2}{N}$$

⇒ So we will have D means
D variances

$$\Rightarrow \left(\frac{x_i^0 - u_i}{\sigma_i}\right)$$

Step 2 ⇒

$$
\begin{array}{c|cccc}
 & & & & \\
x^1 \rightarrow & x_1^1 & x_2^1 & & x_N^1 \\
x^2 \rightarrow & x_1^2 & x_2^2 & & \vdots \\
x^3 \rightarrow & \vdots & \vdots & & \vdots \\
\rightarrow & x_1^D & x_2^D & & x_N^D
\end{array}
$$

$$
C \Rightarrow \begin{bmatrix} C_{x_1x_1} & C_{x_1x_2} \\ C_{x_2x_1} & C_{x_2x_2} \end{bmatrix}_{2\times 2}
$$

Covariance matrix is symmetric

final Covariance matrix ⇒

$$
C = \begin{bmatrix} C_{x_1x_1} & C_{x_1x_2} & & C_{x_1x_D} \\ C_{x_2x_1} & C_{x_2x_2} & \cdots & \\ \vdots & & & \\ C_{x_Dx_1} & \cdots & & C_{x_Dx_D} \end{bmatrix}_{D\times D}
$$

Step 3 ⇒

$$\left(C \underset{\sim}{X} = \lambda X \right)$$

$$\begin{bmatrix} x^1 \\ x^2 \\ x^3 \\ \vdots \end{bmatrix}$$

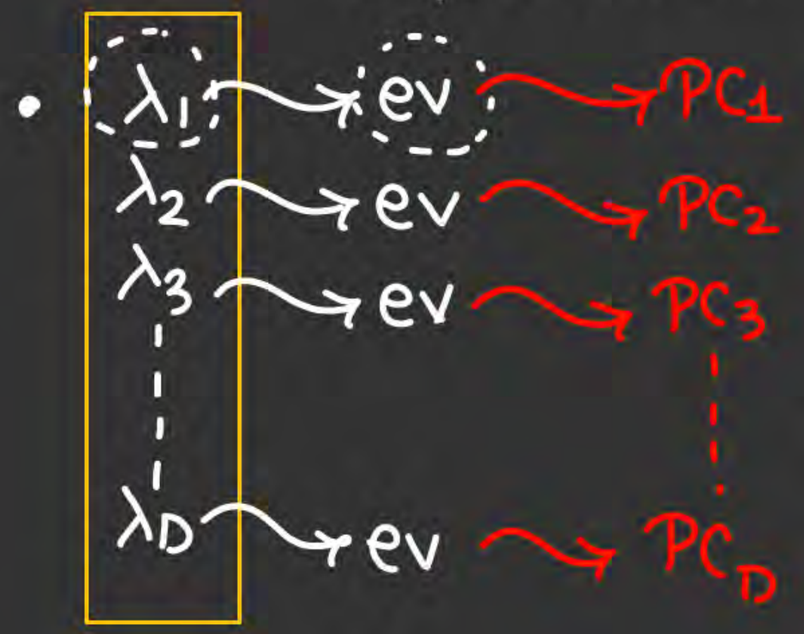
eigen value

eigen vector

If C is a DxD matrix D eigen values
D eigen vectors

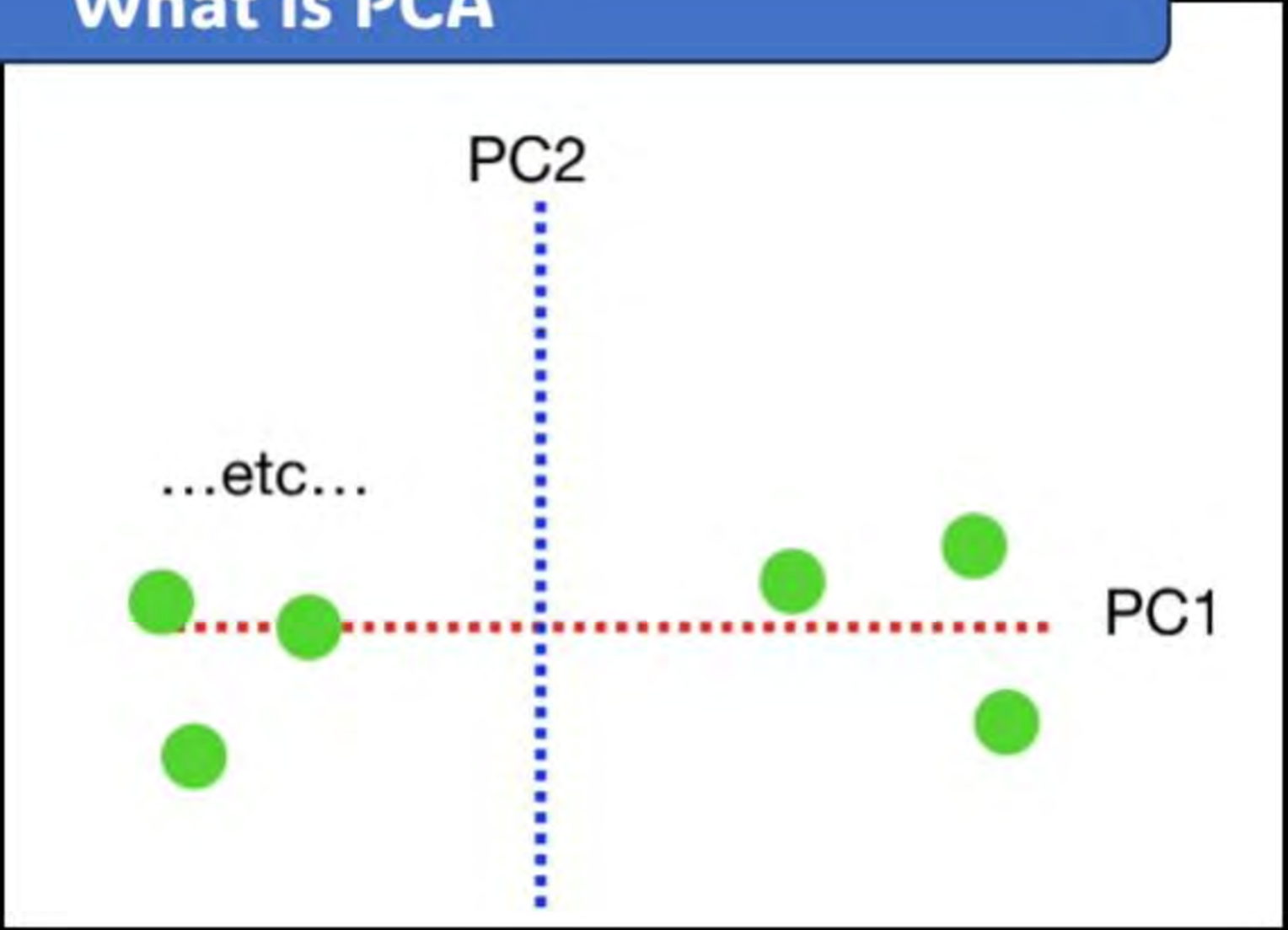
- we write eigen values in descending order
- larger the eigen value more is the information of data collected by eigen vector corresponding to it.

eigen Vectors



- information captured will dec
- So we choose few of these
- these PC's are orthogonal to each other. becoz the matrix is symmetric.

What is PCA



Steps to Perform PCA ...

Step No 1 Standardize the Data:

- Center the data by subtracting the mean of each variable.
- Standardize the data (optional, but recommended) to have unit variance.

Step No 2 Compute the Covariance Matrix:

- Calculate the covariance matrix of the standardized data.
- **Calculate Eigenvalues and Eigenvectors:**
- Determine the eigenvalues and eigenvectors of the covariance matrix.

Step No 3 **Sort Eigenvalues and Eigenvectors:**

- Sort the eigenvalues in descending order and arrange the eigenvectors accordingly.

Step No 4 Select Principal Components: D Dimension → K Dimension

- Choose the top k eigenvectors corresponding to the k largest eigenvalues to form the new feature space.

Step No 5 Transform the Data:

- Project the original data onto the new feature space using the selected principal components

Principal Component Analysis

Step No 1

5D data → λ_1 λ_2 λ_3 λ_4 λ_5

5 eigen values

$\lambda_1 \rightarrow PC_1$, $\lambda_2 \rightarrow PC_2$, $\lambda_3 \rightarrow PC_3$, $\lambda_4 \rightarrow PC_4$, $\lambda_5 \rightarrow PC_5$

$\lambda_1 \rightarrow$ % Capture $\dfrac{\lambda_1}{\text{Total } \lambda\text{'s}}$

$\lambda_2 \rightarrow$ % Capture $\dfrac{\lambda_2}{\text{Total } \lambda\text{'s}}$

LDA V/S PCA

- **Principal Component Analysis (PCA):**
- An unsupervised dimensionality reduction technique.
- Focuses on maximizing variance and identifying the directions (principal components) that capture the most variability in the data.
- **Linear Discriminant Analysis (LDA):**
- A supervised dimensionality reduction technique.
- Focuses on maximizing the separation between multiple classes by finding the linear combination of features that best separate the classes.

and minimize within class variance

LDA V/S PCA

1. **Supervision**:
   - **PCA**: Unsupervised; does not consider class labels.
   - **LDA**: Supervised; uses class labels to maximize class separability.
2. **Objective**:
   - **PCA**: Maximizes the variance in the data.
   - **LDA**: Maximizes the ratio of between-class variance to within-class variance.
3. **Application**:
   - **PCA**: Used primarily for exploratory data analysis, noise reduction, and feature extraction.
   - **LDA**: Used primarily for pattern recognition, classification, and feature extraction in the context of classification tasks.

LDA V/S PCA

- Conclusion

- PCA is more general and can be used for any dataset to reduce dimensionality and explore data structure.
- LDA is more specific and powerful for classification tasks, as it considers class labels to maximize class separation.

Steps

NPoints
100 Dimension data → PCA →

(10Dimension Npoints) ⇒ LR, SVM

now.

ylabel attach

dabel detach from data

PCA – Practise

1. **What is the main goal of Principal Component Analysis (PCA)?**

- A. To classify data points into distinct groups
- B. To maximize variance and identify patterns in the data
- C. To minimize variance in the dataset
- D. To maximize the number of features in the dataset

PCA – Practise

2. **Which of the following is the first step in performing PCA?**

- A. Calculating the covariance matrix
- B. Standardizing the data
- C. Calculating eigenvalues and eigenvectors
- D. Projecting data onto principal components

PCA – Practise

3. **What does each principal component represent in PCA?**

- A. A linear combination of all the original variables
- B. An independent variable
- C. A nonlinear transformation of the data
- D. A class label

PCA – Practise

4. How are the principal components ordered?

By λ's

- A. By the mean of the original variables
- B. By the total number of data points
- C. By the amount of variance they explain
- D. By their correlation with the original variables

PCA – Practise

5. Which matrix is used to determine the directions of the principal components?

- A. Correlation matrix
- B. Covariance matrix
- C. Distance matrix
- D. Identity matrix

PCA – Practise

6. **In PCA, what are eigenvalues used for?**

- A. To measure the direction of principal components
- B. To standardize the data
- C. To measure the variance explained by each principal component
- D. To calculate the mean of the data

PCA – Practise

7. What is the purpose of standardizing the data before performing PCA?

- A. To ensure that the data points are ~~normally distributed~~
- B. To make the data points positive
- C. To ensure each feature contributes equally to the analysis
- D. To increase the dimensionality of the data

PCA – Practise

8. Which of the following is true about the principal components in PCA?

- A. They are always orthogonal to each other
- B. They are always parallel to each other
- C. They have the same magnitude
- D. They are dependent on the number of classes in the data

PCA – Practise

9. **What does a scree plot in PCA help you determine?**

- A. The number of classes in the data
- B. The optimal number of principal components to retain
- C. The mean and variance of the original variables
- D. The standard deviation of the data

PCA – Practise

10. Which of the following best describes the transformation performed by PCA?

$\Rightarrow PC_1 \Rightarrow (x_1 - x_2 = 0)$

- A. Nonlinear transformation
- B. Logarithmic transformation
- C. Linear transformation ✓
- D. Polynomial transformation

Numerical Exam

	X1	X2
1.	2	4
2.	3	5
3.	4	6
4.	5	7
5.	6	8

Step 1 ⇒ $\bar{X}_1 \Rightarrow 2+3+4+5+6/5 \Rightarrow 4$

$\bar{X}_2 \Rightarrow 4+5+6+7+8/5 \Rightarrow 6$

$$\sigma_{X_1}^2 \Rightarrow \frac{\sum(X1-4)^2}{5} = 2$$

$$\sigma_{X_2}^2 \Rightarrow \frac{\sum(X2-6)^2}{5} = 2$$

Principal Component Analysis

Numerical Exam

	X1	X2
1.	2 → $\frac{2-4}{\sigma} \Rightarrow -2/\sqrt{2}$	4 → $4-6/\sqrt{2} \Rightarrow \frac{-2}{\sqrt{2}}$
2.	3 → $3-4/\sigma \Rightarrow -1/\sqrt{2}$	5 → $5-6/\sqrt{2} \Rightarrow -\frac{1}{\sqrt{2}}$
3.	4 → $4-4/\sigma \Rightarrow 0$	6 → 0
4.	5 → $5-4/\sigma \Rightarrow 1/\sqrt{2}$	7 → $\frac{1}{\sqrt{2}}$
5.	6 → $6-4/\sigma \Rightarrow 2/\sqrt{2}$	8 → $2/\sqrt{2}$

Step 1 ⇒ $\bar{X}_1 \Rightarrow 2+3+4+5+6/5 \Rightarrow 4$

$\bar{X}_2 \Rightarrow 4+5+6+7+8/5 \Rightarrow 6$

$$\sigma_{X_1}^2 \Rightarrow \frac{\sum(X1-4)^2}{5} = 2$$

$$\sigma_{X_2}^2 \Rightarrow \frac{\sum(X2-6)^2}{5} = 2$$

x^1	x^2
$-2/\sqrt{2}$	$-2/\sqrt{2}$
$-1/\sqrt{2}$	$-1/\sqrt{2}$
0	0
$1/\sqrt{2}$	$1/\sqrt{2}$
$2/\sqrt{2}$	$2/\sqrt{2}$

unit variance

$$C = \begin{bmatrix} C_{x_1x_1} & C_{x_1x_2} \\ C_{x_2x_1} & C_{x_2x_2} \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\sigma_{x_1}^2 \Leftarrow C_{x_1x_1} \Rightarrow \frac{\sum (x_1-\mu_1)(x_1-\mu_1)}{5} \Rightarrow \frac{\sum x_1^2}{5}$$

$$\Rightarrow \frac{\frac{4}{2}+\frac{1}{2}+0+\frac{1}{2}+\frac{4}{2}}{5}$$

$$\Rightarrow \frac{10}{10} = 1$$

Eigen Values

$$|C-\lambda I| = 0$$

$$\left|\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right| = 0$$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - 1 = 0$$

$$(1-\lambda)^2 = 1$$

$$1-\lambda = \pm 1$$

$$1-\lambda = 1 \quad , \lambda = 0$$

$$1-\lambda = -1 \quad , \lambda = 2$$

$$[C-\lambda I][X]=0 \qquad [X]=\begin{bmatrix} x^1 \\ x^2 \end{bmatrix}$$

Eigen Vectors

$\lambda=0$

$$(C-\lambda I)(X)=0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x^1 \\ x^2 \end{bmatrix}=0 \Rightarrow x^1+x^2=0 \rightarrow PC_2$$

$\lambda=2$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}-\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}=\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x^1 \\ x^2 \end{bmatrix} \Rightarrow x^1-x^2=0 \rightarrow PC_1$$

So PC's are

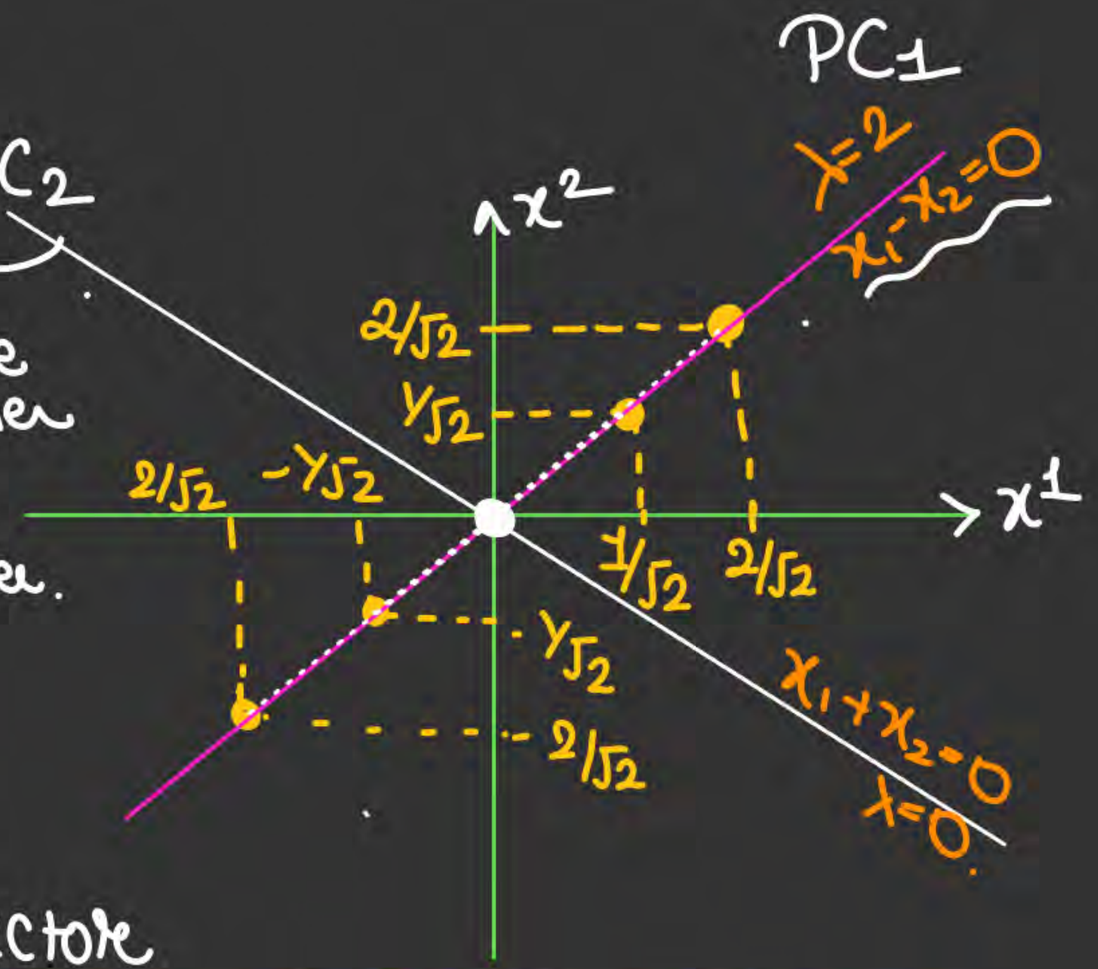
$\Rightarrow$

$\lambda = 2 \Rightarrow x_1 - x_2 = 0$

$\lambda = 0 \Rightarrow x_1 + x_2 = 0$

PC_2 $\leftarrow$ no info/ variance gather

$\rightarrow$ max info gather.



$\Rightarrow$ the eigen value of eigenvector is equivalent to the distance of the projected points on the line from origin

So larger the eigen value ⇒ more the variation of data captured by eigen vector

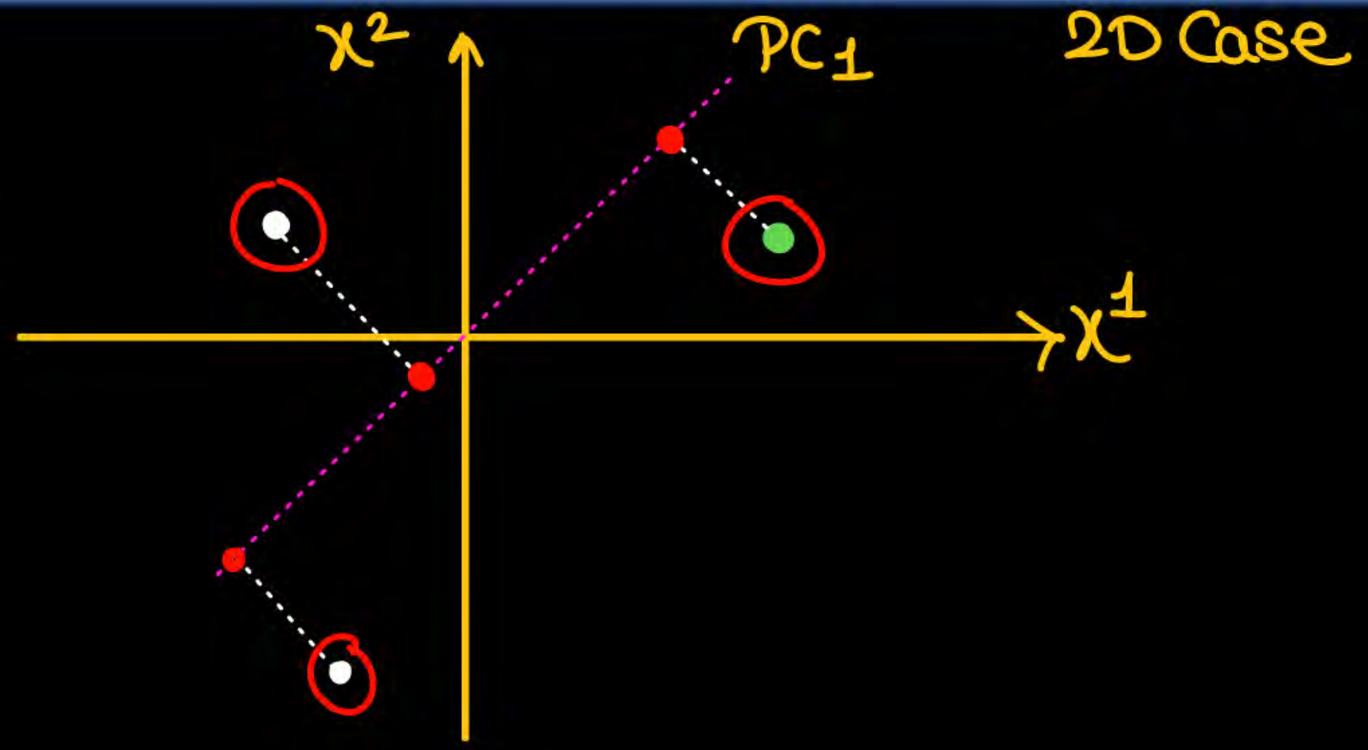
$\lambda_{largest}$ PC_1 K PC's.

$\lambda_{smallest}$ PC_D

Procedure ...

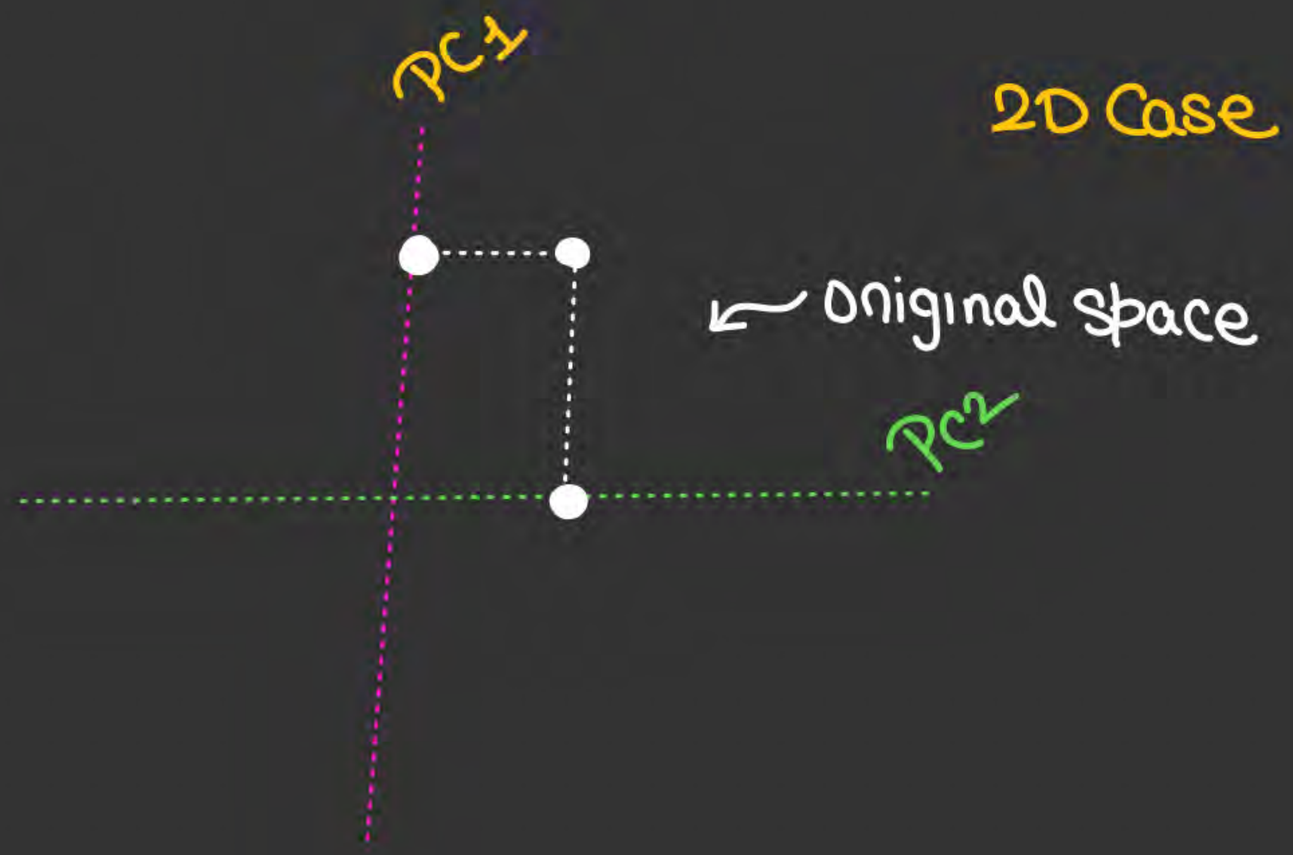
⇒ Single PC_1

$2D \rightarrow 1D$



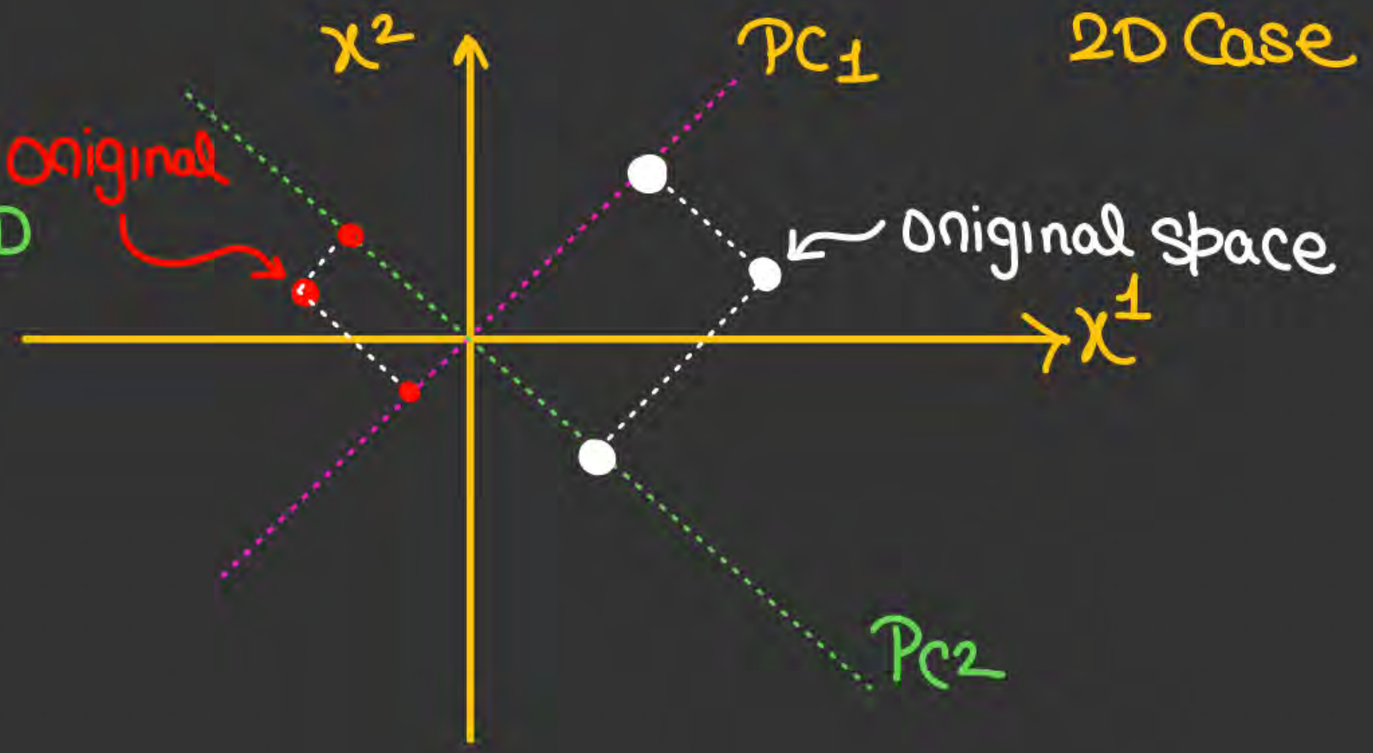
⇒ 2PC's

2D ⇝ 2D



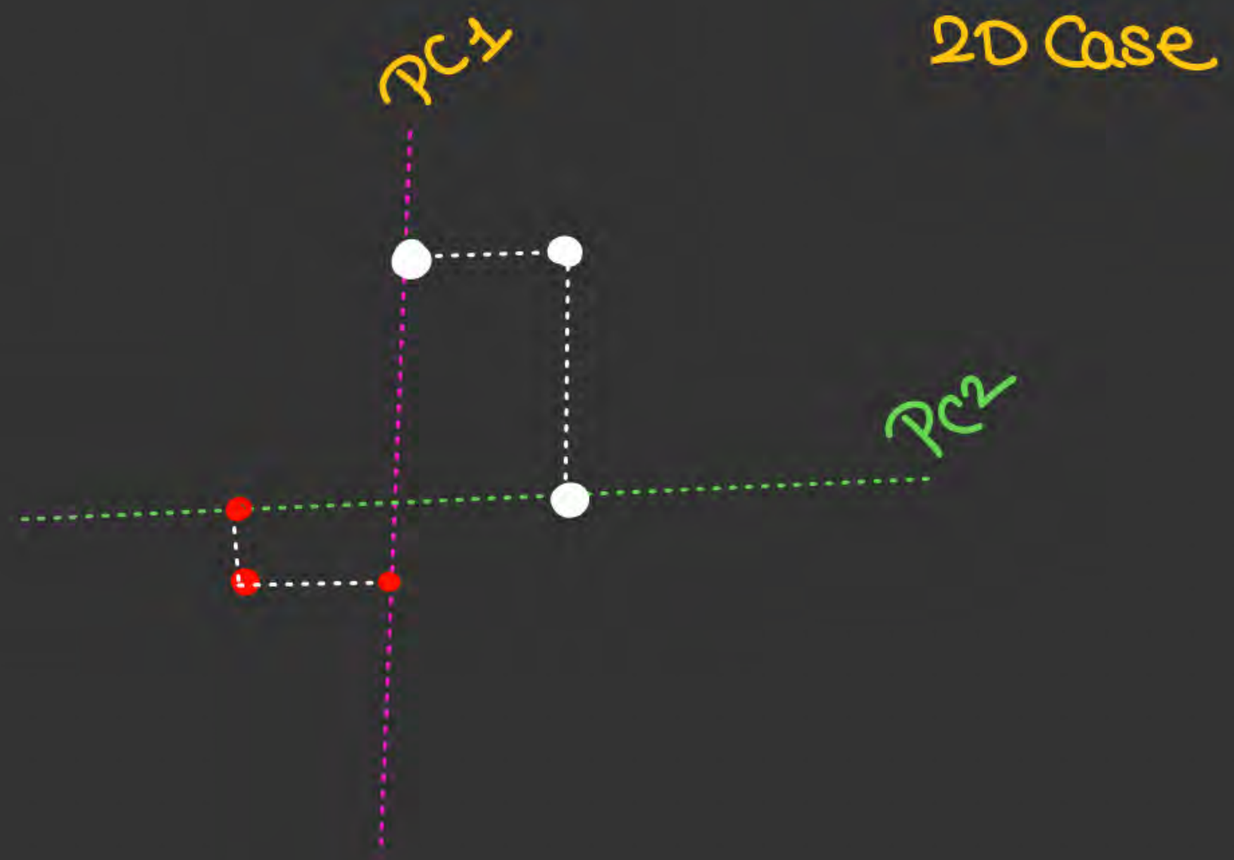
⇒ 2PC's

2D ⇝ 2D



⇒ 2PC's

2D ⇝ 2D



THANK - YOU